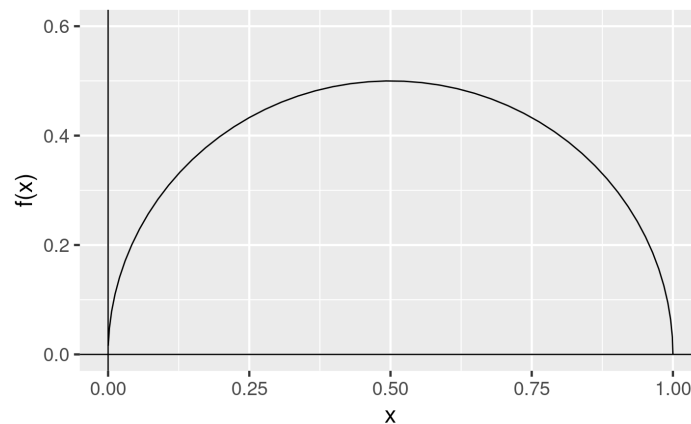


1. True or False: The function below is a valid probability density function. Hint: The area of a circle is πr^2 where r is the radius of the circle.



We are going to study what I would call a “pessimistic random walk”. Say you are in a one-dimensional space, and you can do one of three things, randomly, at any given moment:

- Go forward one step
- Stay put
- Go backwards two steps

2. Let X_1 be the number of steps taken forward/backward in a given moment. Write down $f(x)$, the probability mass function for X_1 .

3. Calculate $\mathbb{E}(X_1)$ and $SD(X_1)$.

4. Suppose you repeat the experiment $n = 10,000$ times, and each decision you make is independent of the last. Calculate $\mathbb{E}(S_{10,000})$ and $SD(S_{10,000})$, as well as $\mathbb{E}(\bar{X}_{10,000})$ and $SD(\bar{X}_{10,000})$.

5. Remember that as n gets large, S_n is approximately $Normal(\mu = E(S_n), \sigma = SD(S_n))$. Find an approximate probability that after 10,000 experiments, you have taken less than 3200 steps total in the “wrong direction.” You may use R to help you calculate this probability.

6. In R, create a vector of 10,000 observations. Each of these observations will be one $S_{10,000} = X_1 + X_2 + \dots + X_{10,000}$. In other words, to obtain one observation, you should sample 10,000 random variables from the distribution of the pessimistic random walk, and take their sum.

7. Now, write R code to create a vector of another 10,000 observations. This time, generate each observation from the approximate distribution of $S_{10,000} = X_1 + X_2 + \dots + X_{10,000}$.

8. Put the two vectors from **Question 6** and **Question 7** into a dataframe called q8. Then, make one histogram of the observations from **Question 6** and another of the observations from **Question 7**. Finally, save each of these histograms into plot objects. Write the R code you used below.

9. Finally, use code from the patchwork library to *stack* the two plots on top of one another. *You need only to copy the patchwork code below.*

10. What should the two distributions look like in relation to one another, and does it match your expectations? If they don’t match your expectations, give a possible reason.

11. Using the column in q8 of the $S_{10,000}$ ’s generated from the **Question 6**, write R code to roughly estimate the probability that you have taken less than 3200 steps in the “wrong direction” after 10,000 repetitions of the experiment that you found in **Question 5**. *Hint: see if you can calculate a proportion.*